

Yu-Chia Chen

Lead LLM Post-training @ Meta (MSL)

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SUMMARY

Staff Research Scientist at Meta Superintelligence Lab (MSL) specializing in NLP and LLM post-training, leading safe, aligned GenAI products from concept to launch.

- Drove measurable global growth by double-digit user engagement grows for AI Characters chatbot
- Delivered tens of million impact and achieved a +6.7% performance gain for LLM-powered ads text generation, with results featured in TechCrunch (yuchaz.github.io/press/202305-techcrunch/) and Import AI (yuchaz.github.io/press/202508-importai/)

TECHNICAL SKILLS

Expertise: LLM; NLP; Reinforcement Learning (RL) post-training (Reward modeling/RLHF/RLUF); Technical Leadership (TL)

Programming: Python: PyTorch, NumPy, Pandas (Advanced); SQL: Presto, Spark (Advanced); C++ (Intermediate)

EXPERIENCE

Staff Research Scientist

Meta

Sep. 2021 – Present

Seattle, WA

Directed multilingual expansion for AI Characters via first-party social data post-training @ MSL

- Achieved double-digit user engagement growth through culturally ad aptive post-training on first-party social conversational data using annealed NLL-PPO for safety-aligned engagement
- Scaled to 50+ languages by architecting multi-year roadmap using pivot coverage and cross-lingual transfer

Led high-performing ads generation via large-scale RL from user/ranking feedback (RLUF) @ Monetization GenAI [1]

- Delivered tens of million impact by leading a two-year, cross-org personalization and multi-agent LLM post-training roadmap
- Achieved +6.7% performance gain by architecting MORL post-training framework, co-optimizing ad performance and brand value
- Grew adoption more than doubled by enabling automated opt-in flow for advertisers through vLLM-based hallucination guardrail

Graduate Research Assistant

University of Washington: Geometric Data Analysis Group

Apr. 2017 – Aug. 2021

Seattle, WA

- Pioneered spectral clustering generalization to higher-order topological features (*NeurIPS'21* oral, <1% acceptance rate) [2]
- Identified and analyzed a key failure mode in spectral embedding methods for complex manifolds (*NeurIPS'19* [7])

Machine Learning Intern

Facebook

Jun. 2020 – Sep. 2020

Seattle, WA

- Optimized search ads recommendation system (recsys) via multi-task learning, achieving Normalized Entropy (NE) gain

Research Intern

Microsoft Research (MSR)

Jun. 2018 – Sep. 2018

Redmond, WA

- Developed scalable dynamic social networks models with causal intervention framework (*KDD'19* [8])
- Partnered with AI for Good Lab on cross-disciplinary wildlife social network analysis (*Animal Behaviour* [3])

EDUCATION

University of Washington

Ph.D. in Electrical Engineering

Seattle, WA

Sep. 2016 – Aug. 2021

National Taiwan University

B.S. in Physics

Taipei, Taiwan

Sep. 2011 – Jun. 2015

PUBLICATIONS (SELECTED)

- [1] Daniel R. Jiang, Alex Nikulkov, **YU-CHIA CHEN**, Yang Bai, and Zheqing Zhu. Improving Generative Ad Text on Facebook using Reinforcement Learning. *arXiv:2507.21983 [cs.LG]*, July 2025
- [2] **YU-CHIA CHEN** and Marina Meila. The decomposition of the higher-order homology embedding constructed from the k-Laplacian. *Advances in Neural Information Processing Systems*, 34, 2021; **NeurIPS oral presentation** (acceptance rate <1%)
- [3] Juan M. Lavista Ferres, Derek E. Lee, Md Nasir, **YU-CHIA CHEN**, Avleen S. Bijral, Fred B. Bercovitch, and Monica L. Bond. Social connectedness and movements among communities of giraffes vary by sex and age class. *Animal Behaviour*, 180:315–328, October 2021
- [4] **YU-CHIA CHEN**. *Learning Topological Structures and Vector Fields on Manifolds with (Higher-Order) Discrete Laplacians*. PhD thesis, University of Washington, 2021
- [5] Samson Koelle, Hanyu Zhang, Marina Meila, and **YU-CHIA CHEN**. Manifold Coordinates with Physical Meaning. *arXiv:1811.11891 [stat.ML]*, July 2021. To appear in JMLR.
- [6] **YU-CHIA CHEN**, Marina Meilă, and Ioannis G. Kevrekidis. Helmholtzian Eigenmap: Topological feature discovery & edge flow learning from point cloud data. *arXiv:2103.07626 [stat.ML]*, March 2021
- [7] **YU-CHIA CHEN** and Marina Meilă. Selecting the independent coordinates of manifolds with large aspect ratios. In *Advances in Neural Information Processing Systems 32*, pages 1086–1095, 2019
- [8] **YU-CHIA CHEN**, Avleen S. Bijral, and Juan Lavista Ferres. On Dynamic Network Models and Application to Causal Impact. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, KDD '19, pages 1194–1204, New York, NY, USA, 2019. ACM